"I'm Not Racist, But…": Social Mixing and Discriminatory Narratives in Swimming Pool Reviews Across Germany

Anonymous submission

Abstract

Social mixing is often encouraged to reduce isolation and foster connection. Yet simply putting different groups in the same space rarely leads to interaction. What matters are the everyday experiences people have in shared social settings—how encounters actually feel and play out.

In this paper, we introduce a method to study these micro-level experiences using user-generated data. We focus on public swimming pools in Germany, collecting 711,004 Google Maps reviews across 4,626 facilities. Using a large language model (Claude-3.5-Sonnet), we identify social encounters and discourse related to race, ethnicity, culture, or religion. We then link these reviews to demographic and political data from 299 German constituencies to explore how racialised discourse varies across space.

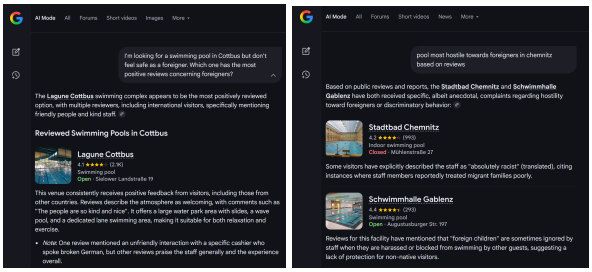
Our findings show that user-generated reviews can reveal the quality of social mixing at the micro and macro levels, offering a way to understand how diversity is experienced in everyday public spaces.

Introduction

Understanding social mixing requires attention to the quality of everyday interactions, not just the presence of different groups in the same place (Amin 2002; Watson 2006). However, studying these interactions presents ongoing methodological challenges. Ethnographic research offers rich, detailed insights but is difficult to extend beyond a small number of sites (Apelt et al. 2023). Quantitative studies based on administrative data, by contrast, often overlook how people experience inclusion and exclusion in daily life, which is central to evaluating social integration.

We address this gap by developing a computational framework that uses user-generated content—specifically, Google Maps reviews—to examine intergroup dynamics at scale. These voluntary, context-specific reviews capture how people experience inclusion, exclusion, and conflict, which are central to how social mixing feels in everyday life. Although reviews do not describe objective conditions, they capture the quality of encounters in ways that traditional data sources cannot. Beyond its research value, this content is increasingly influential: it informs recommendation systems and large language models that shape how people access urban facilities (Bender et al. 2021). This creates feedback loops in which reported social tensions affect future patterns of use without direct human oversight (Figure 1).

We validate and demonstrate our approach by analysing public swimming pools across Germany. Pools serve as an ideal empirical testbed for three reasons. First, they are intimate micro-settings in which bodies, norms, and identities are visibly regulated, thereby amplifying the stakes of intergroup contact (Waller Ph D and Bemiller JD 2018). Historically, they are sites of violent exclusion around race, class, and gender; they remain arenas where secular norms, religious practices (e.g., Muslim swimwear), and institutional rules collide (Altmeyer and Weber 2021). Second, the German context provides rich structural variation: distinct migration histories (from Gastarbeiter to recent refugees), federal diversity in integration policies, and ongoing public controversies over religious accommodation (Michalowski 2021).



(a) Cottbus (b) Chemnitz

Figure 1: AI-generated summaries in a search engine drawing on Google Maps reviews when recommending public swimming pools to a non-local user in Cottbus and Chemnitz, cities in Germany

Following research on racialization, we use the term *racial discourse* to denote the language through which social boundaries are constructed, naturalised, or contested along axes of perceived difference—including race, ethnicity, religion, and culture—regardless of whether biological race is explicitly invoked. We pursue these questions through a multi-stage analytical pipeline combining automated classification, predictive modelling, and qualitative interpretation to address three research questions:

- Does user-generated content capture racial discourse in

micro-settings, and what does it look like?

- What are the social themes and linguistic patterns that characterise racial discourse?
- Can structural factors (pool attributes, constituency demographics, political orientation) be used to predict the spatial variation of racial discourse?

It is worth noting that there are some limitations to user-generated reviews. Most critically for social-mixing purposes, reviews documenting social friction may reflect prejudice rather than describe objective conditions. We treat this not as a fatal flaw but as an interpretive opportunity. Reviews do not provide ground truth about integration; they reveal how integration is subjectively experienced and publicly narrated. The presence of racial discourse in reviews is itself sociologically meaningful: it indicates sites where difference is salient, norms are challenged, and people feel able to publicly draw social boundaries. Our work introduces a scalable framework for extracting encounter quality from digital traces, demonstrating how computational analysis can complement traditional ethnographic approaches. Substantively, we provide the first systematic, multi-city analysis of intergroup dynamics in German public pools, revealing patterned variation in how diversity is negotiated across intimate leisure spaces. While demonstrated here through German public swimming pools, our framework is transferable to other contested semi-public spaces (e.g., libraries, community centres, parks) and to national contexts where user-generated content documents everyday encounters across difference.

Related Work

Understanding social integration requires moving beyond spatial co-presence to examine the quality of everyday interactions (Amin 2002; Watson 2006). Urban research shows that simply sharing space rarely produces meaningful contact (Fincher et al. 2014; Lees, Slater, and Wyly 2013). Yet studying encounter quality at scale remains methodologically challenging. Ethnographic work offers rich contextual insight but is difficult to generalise (Apelt et al. 2023), while surveys and administrative data capture broader patterns at the expense of lived experience. As a result, large-scale analyses of intergroup dynamics remain limited.

Research on micro-level integration has largely relied on three methodological traditions, each with clear constraints. Ethnographic studies document how inclusion and exclusion are negotiated in specific settings such as swimming pools, community centres, and public parks (Apelt et al. 2023; Waller Ph D and Bemiller JD 2018; Altmeyer and Weber 2021). However, their depth comes at the cost of scale, as intensive fieldwork typically spans only a handful of sites. Survey-based approaches enable larger samples but face validity challenges: retrospective self-reports are prone to recall and social desirability bias and tend to measure contact frequency rather than encounter quality (Pettigrew and Tropp 2006; Amin 2002). Administrative data on residence, schooling, or employment capture aggregate patterns of mixing but reveal little about how encounters are experienced—whether as meaningful interaction, superficial co-presence, or active avoidance.

User-generated reviews from location-based online services offer an emerging alternative that may bridge these trade-offs. Reviews are voluntary, situated, and available at scale, capturing evaluative dimensions such as comfort, frustration, and perceived (in)civility that shape engagement with mixed spaces. Despite the growing availability of such data, their potential for studying social integration remains underexplored. Existing research has focused primarily on economic outcomes, platform dynamics, or sentiment analysis for business intelligence (Luca 2016; Siles et al. 2019), with less attention to reviews as windows into the micro-politics of public space—how diversity is noticed, narrated, and evaluated in specific places.

Our methodological contribution is the development of a systematic framework for collecting and analysing Google Maps reviews to study place-specific intergroup dynamics at scale.

We validate this framework using public swimming pools across Germany, a setting that provides a rigorous methodological test for three reasons. First, pools offer strong theoretical leverage. Although urban policies promoting social mixing often draw on contact theory (Pettigrew and Tropp 2006), research documents a persistent implementation gap: initiatives such as mixed-income housing or neighbourhood renewal frequently produce tokenistic mixing that obscures displacement (Fincher et al. 2014; Lees, Slater, and Wyly 2013). German case studies show that spatial proximity alone can reinforce symbolic boundaries rather than dissolve them (Huning and Schuster 2015; Matejskova and Leitner 2011). Pools differ from residential settings in that visitation is voluntary, making them a revealing context for observing when and how integration succeeds or fails.

Second, pools heighten the visibility of intergroup dynamics. As intimate leisure spaces where bodies, norms, and identities are explicitly regulated, they intensify the stakes of contact (Waller Ph D and Bemiller JD 2018). Historically, pools have been sites of racial exclusion, class stratification, and gender regulation, and they remain arenas where secular norms and religious practices—such as Muslim swimwear—come into tension (Waller Ph D and Bemiller JD 2018; Altmeyer and Weber 2021; Michalowski 2021). Prior work shows that these same features allow pools either to facilitate meaningful contact or to reproduce exclusion through spatial design and informal regulation (Apelt et al. 2023).

Third, Germany provides the structural variation required for model validation. Segregation patterns reflect layered migration histories, from Gastarbeiter recruitment to post-reunification mobility and recent refugee arrivals (Meister and Niebuhr 2021). These trajectories produce group-specific inequalities, including heightened residential segregation among Turkish-origin populations (Wolter, Raviv, and Mertens 2023) and ethnic "halo effects" extending beyond neighbourhood boundaries (Jünger and Schaeffer 2023). Federal variation in integration policies, urban form, and demographic composition across cities creates a natural testbed for assessing whether our method (RQ3) recovers

known structural patterns.

Existing research on swimming pools remains fragmented: qualitative analyses of accommodation policies (Michalowski 2021) coexist with small-N ethnographies (Apelt et al. 2023), but systematic investigation of user experiences across cities is absent. German pools thus constitute an ideal empirical testbed—conceptually rich, empirically underexplored, and offering the variation needed to validate scalable approaches.

In sum, while research shows that encounter quality matters more than spatial proximity, scalable methods for examining lived experience across contexts remain limited. We address this gap with a computational framework that preserves interpretive depth while enabling systematic comparison. German pools provide the empirical variation needed to validate this approach and to reveal how diversity is navigated in contested leisure spaces.

Data & Methods

Methodological Framework

We use the following framework in our data collection and analysis:

1. **Corpus Construction**: Through our pipeline, we collect Google Map reviews on public swimming pools in Germany and geo-code them to administrative data.
2. **Data Classification**: We then employ a multi-stage classification and validation process to identify reviews containing racial discourse and their intersecting social themes.
3. **Predictive Analysis**: Finally, we predict the spatial variation of racial discourse across Germany.

Data Collection Pipeline

We define *public swimming pools* as aquatic leisure infrastructure accessible to the public upon payment of standard fees, excluding facilities that require club membership or hotel accommodation. This definition prioritises **voluntary leisure sites** in which use constitutes an act of self-selection—a key theoretical property for studying subjective integration experiences.

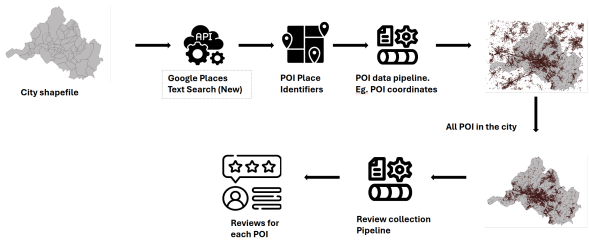


Figure 2: Caption

Using an established Google Maps collection pipeline (Mensah, Cannanure, and Weber 2025), we collect reviews on public swimming pools from Google Maps. Figure 2 shows a general implementation of our data collection pipeline. We then implemented two-stage geographic matching of data, coordinate proximity within 500 meters and address-string matching for disambiguation, to a crowdsourced inventory of pools on the Baederleben.de website (Bäderleben 2024). We retained $4,296$ validated pools (83% of Baederleben reference set) plus 898 Google Maps-only facilities, yielding **4,837 pools**. The gaps primarily reflect seasonal closures during data collection (August–October 2025) on Google and category misclassifications in the reference database.

Data collection (August–October 2025) for the $4,837$ pools yielded **2,378,782** total reviews and ratings, **1,111,331** textual and this covered (46.7%) $4,714$ pools. The data contains several review attributes. We use only the following attributes: *total reviews, per-pool rating, review_id, reviewer name, reviewer rating, count of other reviews on Google by the reviewer, relative timestamps, and review text (English translation)*. Figure 6 in the appendix shows the uncertainty-aware review distribution. Google's relative timestamps (e.g. "3 years ago", "a month ago") preclude exact dating, but bounded intervals show coverage from 2015–2025, with concentration in recent years.

We use Acua (Jung, Salminen, and Jansen 2020), a name-to-ethnicity mapping tool, to predict the likely country of origin as a proxy for reviewers' ethnic backgrounds (Rabiei-Dastjerdi, McArdle, and Aghajani 2022). While imperfect (subject to naming conventions and migration patterns), this enables aggregated demographic analysis of *who* participates in racial discourse—a dimension inaccessible via review text alone. We geocoded our data to constituency-level demographic and electoral data we obtained from the German Federal Returning Office website (Der Bundeswahlleiter 2023). This enables constituency-level analysis of how local political and demographic contexts correlate with review discourse (RQ3).

Data Classification

User reviews of places generally range from vague to specific, and a single review can address multiple topics. Swimming pool reviews are no different. Additionally, the type of discourse we aim to identify is expected to be rare, subtle, and implicit in the statement's wording, given Google's platform regulations. As such, subjective and context-aware judgment is required to distinguish from superficially similar content. Standard binary classification (e.g., "racist" vs. "not racist") risks conflating distinct phenomena and misses meaningful gradations.

We created a **three-stage hierarchical classification schema** for identifying racial discourse, which is adaptable to studying social mixing in other settings. First, we identify whether a review describes a *social encounter*, filtering out purely infrastructural feedback while retaining observations involving staff, other patrons, or the social atmosphere. This step ensures that subsequent analysis is grounded in interactional contexts. Second, social encounter reviews are screened for *racialisation*, defined broadly to include references to race, ethnicity, culture, or religion. Consistent with a constructivist view of racialisation (Omi and Winant 2014), we include both explicit markers and implicit boundary-making. Third, racialised reviews are cate-

gorized by connotation as *discriminatory*, *pleasant*, or *ambiguous*. Distinguishing these categories allows us to assess encounter quality rather than treating all racial references as equivalent. Table 10 in the appendix explains the schema in more detail and provides illustrative examples for each classification step.

Annotation and Model Selection

The authors randomly sampled reviews and manually screened them, then used a keyword dictionary to filter the review corpus for potential racially biased reviews. From this set a random sample of 100 reviews were selected. Four annotators independently labelled these 100 reviews using the hierarchical classification schema (30 reviews with discussion). Krippendorff's α and Fleiss' κ exceeded 0.73 across all dimensions —above the 0.67 reliability threshold (Krippendorff 2011). (See Table 9 in the appendix). We created consensus labels for 100 reviews using majority voting among four trained annotators, with expert adjudication for ties. This produces a high-confidence validation set reflecting collective human judgment. We evaluated eight state-of-the-art LLMs and two additional human annotators against the ground truth using identical prompts derived from Table 10 for the same set of 100 reviews.

Model/Annotator	Acc.	Prec.	Rec.	F1
Claude-3-Haiku	0.820	0.852	0.820	0.824
Claude-3.5-Sonnet	**0.937**	**0.932**	**0.937**	**0.933**
DeepSeek-R1	0.937	0.941	**0.937**	0.938
DeepSeek-Chat	0.917	0.921	0.917	0.917
Gemini-2.5-Flash	0.930	0.931	0.930	0.929
Gemini-2.5-Pro	0.880	0.893	0.880	0.878
GPT-4o	0.900	0.906	0.900	0.901
GPT-4o-Mini	0.897	0.909	0.897	0.899
Human 1	0.910	0.924	0.910	0.912
Human 2	0.877	0.885	0.877	0.879

Table 1: Model performance comparison for social discourse classification.

Table 1 shows the models and their performance. Based on performance and cost considerations, we selected `Claude-3.5-Sonnet` for full corpus classification. To manage API costs, reviews exceeding 200 characters were summarised using `philschmid/bart-large-cnn-samsum` before classification.

Classified Reviews

711,004 reviews (64% of the textual corpus) were successfully classified by the model, representing 98.1% $(4,626)$of pools . There were a few instances in which the model treated the labels individually rather than hierarchically. Given the logical flow of our classification schema, this was easy to correct. Due to financial constraints, we did not reclassify the remaining reviews; instead, we assessed whether the classified set was sufficiently representative to proceed with our analysis. We check that the classified subset is spatially representative and free of systematic volume bias (see Figure 7 in the Appendix). All subsequent analysis is based on this classified subset.

To identify social interaction themes in the classified racial reviews, we again employ the `Claude-3.5-Sonnet` model. First, we prompt it to discover 10 to 15 recurring themes in the reviews by passing the reviews in batches of 100. The discovered themes were aggregated using a second prompt. A final set of 11 themes was then used to label the racial reviews via a third prompt. To validate the model's classification, we select a stratified random sample of the labelled reviews.

Results and Findings

Table 2 provides a summary of the results from our classification of reviews. 36.4% of the classified $711,004$ reviews described or contained a description of a social encounter. This shows that even in places with a high likelihood of intergroup contact, users leave more reviews on topics (e.g., price, water temperature) that do not involve contact with others. As expected, the racial discourse is rare but present. Our model identifies less than 1% of the entire dataset as containing racial discourse, which is just below 2% of reviews with social encounters. While this does not equal social cohesion, in such intimateof pools, it is encouraging. The Nürnberg-Süd constituency in the Bayern state is the only one for which no racial reviews are identified. It has 2 pools in the classified dataset, one with 325 reviews and the other with 1.The constituency's review count remains the same even in the larger corpus of 1.1 million reviews; hence, this is not an artefact of using the subset of classified data. Racial reviews were spread across $1,491$ pools in 298 constituencies. Geographically, racial discourse does not appear to be concentrated Among connotation types in racial reviews, the discriminatory type accounts for more than half, which is notable. Given that discrimination encompasses both experienced (perceived) and perpetrated forms, and given Google's platform policies, it would be interesting to explore the nature of this connotation and its linguistic use. However,that is not covered in this paper.

We also examine the characteristics of reviewers who left these racial reviews, without conducting any individual-level analysis. The $4,923$ racial reviews were left by $4,923$ reviewers. Of these, we were unable to estimate the nationality of about 408 (8.2%), compared with $48,626$ (6.8%) in the non-racial reviews, which had 567096 unique reviewers. Among the unique reviewers in the racial reviews, $28(0.6\%)$ leave multiple reviews; however, across different pools, the highest number of reviews left by a single reviewer is 3. By contrast, in the non-racial reviews, there are $92615(16.2\%)$ leaving multiple reviews across pools, with the highest reviewer leaving 78 reviews. This suggests that real people are leaving racial reviews, and that these reviews are not coordinated but rather reflect lived experiences.

For the estimated nationalities, Table 4 presents the distribution of the top nationalities across both racial and non-racial reviews. With the exception of Brazil and Indonesia, all others appear in both review types in relatively the same order.

Descriptive Metric	Value
Social Encounters	
Number of reviews	259,026 (*36.4%*)
Racial Discourse	
Reviews with racial content	4,953
% of social encounters	*1.9%*
% of entire corpus	*0.7%*
Connotation Types	
Discriminatory	2,734 (55.2%)
Ambiguous	1,219 (24.6%)
Pleasant	1,000 (20.2%)
Geographic Spread	
Pools with racial discourse	1,491 (*32.2%*)
Constituencies with racial discourse	298(*99.7%*
Distribution Across Constituencies	
Mean racial reviews per constituency	17
Median racial reviews per constituency	12.0
Standard deviation	17.9
Maximum racial reviews	177
Minimum racial reviews	1
Racial Review Proportions	
Mean racial proportion (%)	0.719
Median racial proportion (%)	0.603
Mean racial reviews per 1000 reviews	7.2
Demographic Context	
Mean foreign population (%)	14.02
Correlation (racial proportion vs foreign population)	0.39

Table 2: Prevalence and Distribution of Racial Discourse in Public Pool Reviews in German constituencies

Table 3: Nationality Distribution of Reviewers

	Non-racial	Racial
Total (with data)	662,308	4,543
Missing (%)	6.8%	8.3%
Germany	41.1%	28.5%
United States	13.6%	11.8%
Russian Federation	2.1%	2.7%
Poland	2.0%	2.0%
Italy	1.7%	1.6%
Turkey	1.6%	4.5%
Philippines	1.6%	1.6%
France	1.4%	2.1%
Brazil	1.4%	–
Indonesia	–	2.0%
nan	6.8%	8.3%

Table 4: Estimated nationalities using reviewer names. Dashes indicate ethnicity not in top 10 for that review type.

Social Interaction Themes

Predicting Spatial Variation in Racialised Discourse

Racial discourse appeared in 0.717% of reviews on average (SD = 0.005), ranging from 0% to 3.853% across constituencies (5).

Variable	Mean	SD	Min	Max
Racial discourse rate (%)	0.717	0.500	0.000	3.853
Foreign population (%)	16.23	8.15	3.10	42.90
Population density (inh./km^2)	1,245	1,587	75	12,589
AfD vote share (%)	12.45	6.80	4.20	31.80
Green Party vote share (%)	15.67	5.92	5.10	34.20
Age 25–34 (%)	12.89	2.34	8.10	19.80
Total reviews (constituency)	2,378	1,892	124	11,045

Table 5: Summary statistics for outcome variable and key predictors (N = 299 constituencies)

Bivariate Relationships and Spatial Patterns

Spearman correlation analysis revealed significant relationships between racial discourse rates and structural factors (3). Population density showed the strongest positive correlation ($r = 0.501$, $p < 0.001$), followed by foreign population percentage ($r = 0.446$, $p < 0.001$). Contrary to expectations, AfD vote share correlated negatively with racial discourse ($r = -0.312$, $p < 0.001$), while Green Party vote share showed a positive correlation ($r = 0.307$, $p < 0.001$). Review volume was not significantly associated with discourse rates ($r = -0.085$, $p = 0.144$).

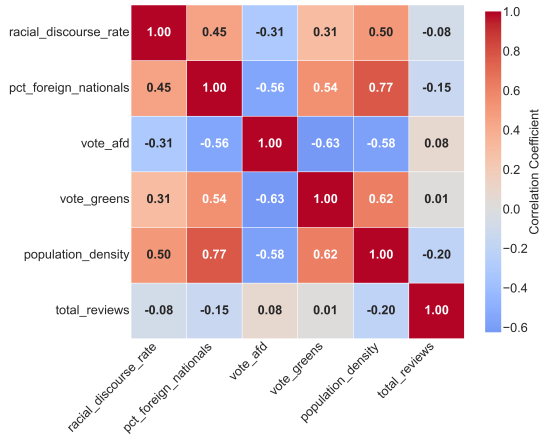
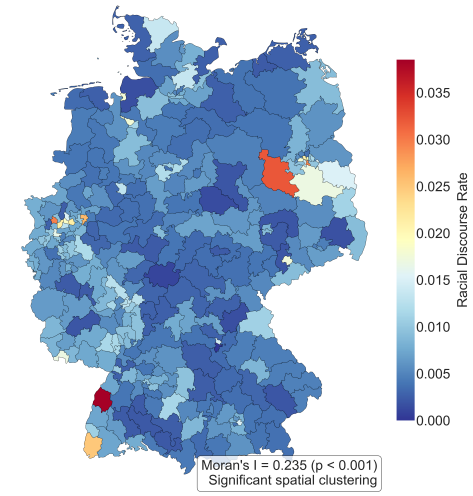
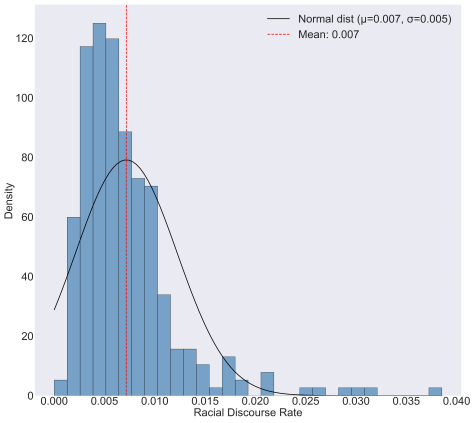


Figure 3: Spearman correlations between racial discourse rates and structural factors. All correlations except total reviews are statistically significant ($p < 0.001$).

Spatial autocorrelation analysis using Moran's I revealed significant geographic clustering of racial discourse ($I = 0.235$, $p < 0.001$), indicating that constituencies with similar discourse rates tend to be geographically proximate. However, the distribution was right-skewed, with most constituencies exhibiting low rates (¡1%) and a long tail of constituencies with elevated rates (**??**).



(a) Geographic Distribution



(b) Statistical Distribution

Figure 4: Geographic distribution of racial discourse across constituencies

Multilevel Modelling and Constituency-Level Analysis

Multilevel logistic regression assessing individual reviews nested within constituencies revealed minimal clustering at the constituency level ($ICC = 0.0034$), indicating that only 0.3% of variance in racial discourse occurred between constituencies. However, constituency-level factors showed significant effects: foreign population percentage ($\beta = 0.047$, $p < 0.001$) and AfD vote share ($\beta = 9.71e-06$, $p < 0.001$) were positively associated with racial discourse when controlling for individual-level factors (reviewer rating, place rating).

Constituency-level OLS regression models further clarified these relationships (6). Demographic factors alone explained 22.3% of variance in racial discourse rates, with population density ($\beta = 0.000$, $p < 0.001$) and foreign population percentage ($\beta = 0.0002$, $p = 0.007$) as significant predictors. Political factors explained only 12.8% of variance, with CDU vote share showing a negative association ($\beta = -0.000$, $p < 0.001$) and SPD vote share a positive association ($\beta = 0.000$, $p = 0.047$). A combined model explained 25.1% of variance, with the percentage of 25-34 year olds emerging as a significant negative predictor ($\beta = -0.0006$, $p = 0.003$).

Predictor	**Model 1 Demographics**	**Model 2 Politics**	**Model 3 Combined**
Constant	0.0056***	0.0072***	0.0195***
Foreign pop. (%)	0.0002**		0.0003***
Pop. density	0.0000***		0.0000***
Total pop.	-0.0000		-0.0000
Age <18 (%)	0.0001		0.0002
Age 25–34 (%)			-0.0006**
AfD vote		0.0000	0.0000
Green vote		0.0000	
SPD vote		0.0000*	
CDU vote		-0.0000***	
Observations	299	299	299
R^2	0.223	0.128	0.251
Adj. R^2	0.212	0.116	0.236
F-statistic	21.10***	10.80***	16.35***

Table 6: OLS regression coefficients predicting racial discourse rates. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

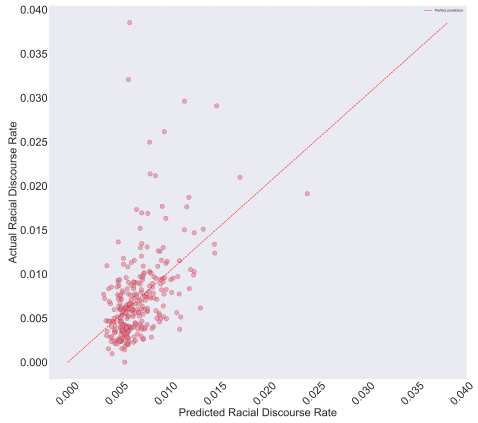
Predictive Modelling and Feature Importance

To assess predictive accuracy, we trained multiple machine learning models on 70% of constituencies ($n = 209$) and tested on a held-out 30% ($n = 90$). Linear regression demonstrated the strongest performance (Test $R^2 = 0.190$, RMSE = 0.0045), with cross-validation confirming robustness (CV $R^2 = 0.2030.093$). Complex algorithms (Random Forest, Gradient Boosting, XGBoost) exhibited severe overfitting, suggesting relationships are primarily linear and additive (7).

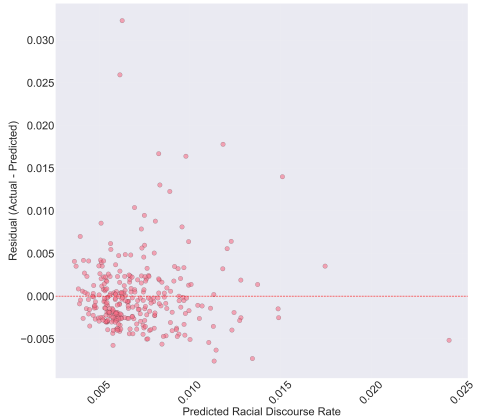
Model	**Train R^2**	**Test R^2**	**Test RMSE**	**Cross-Val R^2**
Linear Regression	0.264	0.190	0.0045	0.203 ± 0.093
Random Forest	0.866	0.069	0.0048	0.062 ± 0.153
Gradient Boosting	0.917	-0.082	0.0052	-0.182 ± 0.519
XGBoost	0.977	-0.239	0.0056	-0.063 ± 0.164

Table 7: Predictive performance on held-out test data (n = 90 constituencies).

Permutation importance analysis revealed a clear hierarchy of predictive factors (**??**). Foreign population percentage emerged as the most important predictor (importance = 0.3440.089), followed by population density (0.2790.060). The percentage of 25-34 year olds showed moderate importance (0.1390.086), while political factors contributed minimally to prediction accuracy (Green Party: 0.0140.009; AfD: 0.0050.004).



(a) Predicted vs. Actual



(b) Residuals

Figure 5: Actual, Predicted and Residuals of racial discourse across constituencies

Identifying Locations for Intervention

Residual analysis revealed systematic prediction errors with important implications for intervention targeting (Figure 4). The model severely underestimated racial discourse in several constituencies, most notably Offenburg (actual: 3.85%, predicted: 0.63%, residual: $+3.22\%$), Brandenburg an der Havel ($+2.60\%$), and Duisburg II ($+1.78\%$). These "crisis areas" exhibited discourse rates 2-6 times higher than model predictions, suggesting unmeasured local factors driving elevated discourse.

Conversely, the model overestimated discourse in constituencies like Berlin-Pankow (-0.76%), München-Nord (-0.73%), and Hamburg-Eimsbüttel (-0.63%), suggesting the presence of unmeasured protective factors. A composite risk score (weighting predicted discourse rates, foreign population percentage, and AfD vote share) identified ten high-risk constituencies, predominantly in Berlin and the industrial Ruhr region (8).

Constituency	Race (%)	For. (%)	AfD Votes
Berlin-Marzahn-Hellersdorf	1.15	21.9	46,285
Berlin-Lichtenberg	0.99	21.6	34,635
Ludwigshafen/ Frankenthal	1.16	22.7	37,871
Gelsenkirchen	1.50	24.3	30,032
Berlin-Pankow	0.37	22.9	31,087
Offenbach	1.05	29.2	26,232
Duisburg II	2.96	25.8	27,428
Heilbronn	0.58	21.7	49,241
Essen II	1.87	24.8	24,911
Pforzheim	0.87	21.2	43,959

Table 8: High-risk constituencies identified by composite risk score.

Discussion

Our analysis yields three principal findings regarding RQ2's focus on prediction, explanation, and intervention sites for racialised discourse.

First, spatial variation in racial discourse is predictable from structural factors, but with important limitations. The 19% of variance explained (Test $R^2 = 0.190$) represents moderate predictive accuracy—sufficient for systematic risk assessment but insufficient for deterministic prediction. This finding aligns with ecological studies of prejudice that identify structural correlates while acknowledging substantial local variation (). The dominance of linear models over complex algorithms suggests additive rather than interactive relationships, simplifying intervention design but potentially missing threshold effects or context-dependent dynamics.

Second, demographic composition—particularly foreign population percentage and population density—provides greater explanatory power than political orientation. This challenges simplistic left-right political explanations of racial discourse and supports urbanisation-based theories of intergroup contact and conflict (). The negative association between young adult populations (25-34) and racial discourse suggests potential generational or lifecycle effects, possibly reflecting differing socialization experiences or online behaviours among younger cohorts ().

Third, model errors provide crucial intervention intelligence beyond structural risk factors. The identification of "crisis areas" with severe model underestimation (e.g., Offenburg at 3.85% discourse) highlights the importance of local factors unmeasured in structural models. These areas represent priorities for immediate investigation and potentially rapid-response interventions. Conversely, "resilient areas" with lower-than-predicted discourse (e.g., Berlin-Pankow) offer opportunities to study protective factors that could be replicated elsewhere.

The spatial clustering of discourse (Moran's $I = 0.235$) coupled with minimal constituency-level variance (ICC = 0.0034) presents an intriguing paradox: while discourse clusters geographically, constituency boundaries explain almost none of this clustering. This suggests that relevant social processes operate at finer scales (neighborhoods, social

networks) or transcend administrative boundaries through media diffusion or regional cultural patterns ().

Ethical Implications & Limitations

Our use of publicly available reviews and aggregated demographic data raises ethical questions and imposes limitations. First, while user data is ubiquitous, it may not necessarily be representative of the typical user population of places of interest.

Conclusion

This analysis demonstrates that our methodological framework can be used to study social mixing and other sociocultural phenomena at the micro level at scale. Our application to public swimming pools in Germany, empirically proves that we geocoding user generated reviews with other spatial data can provide useful insights into geospatial patterns.

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Ethics Checklist

1. For most authors...
   (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? Yes
   (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope? Yes

(c) Do you clarify how the proposed methodological approach is appropriate for the claims made? Yes

(d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? Yes, in methods and discussion

(e) Did you describe the limitations of your work? Yes, together with ethical implications

(f) Did you discuss any potential negative societal impacts of your work? Yes, together with limitations

(g) Did you discuss any potential misuse of your work? Yes, together with ethical implications

(h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? Yes

(i) Have you read the ethics review guidelines and ensured that your paper conforms to them? Yes

2. Additionally, if your study involves hypotheses testing...

(a) Did you clearly state the assumptions underlying all theoretical results? Yes, we have discussed this in the Methods

(b) Have you provided justifications for all theoretical results? Yes, in the discussion

(c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? Yes, in the limitations

(d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? Yes, in the limitations

(e) Did you address potential biases or limitations in your theoretical framework? Yes, in the discussion

(f) Have you related your theoretical results to the existing literature in social science? Yes, in the discussion

(g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? Yes, in the discussion

3. Additionally, if you are including theoretical proofs...

(a) Did you state the full set of assumptions of all theoretical results? NA

(b) Did you include complete proofs of all theoretical results? NA

4. Additionally, if you ran machine learning experiments...

(a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? NA but we used an external tool for name-to-ethnicity classification, which is listed in the methods

(b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? NA

(c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? NA

(d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? NA

(e) Do you justify how the proposed evaluation is sufficient and appropriate to the claims made? NA

(f) Do you discuss what is "the cost" of misclassification and fault (in)tolerance? Yes, we discuss the name to ethnicity classification in the limitation

5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...

(a) If your work uses existing assets, did you cite the creators? Yes, we are using the Google Maps data

(b) Did you mention the license of the assets? Yes, it is free for public use

(c) Did you include any new assets in the supplemental material or as a URL? No.

(d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? NA, publicly available data

(e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? Yes, we mention that this (public) data contains names when a reviewer chooses to provide one

(f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR? NA

(g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset? NA

6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...

(a) Did you include the full text of instructions given to participants and screenshots? NA

(b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? NA

(c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? NA

(d) Did you discuss how data is stored, shared, and deidentified? NA

Appendix

Dimension	**Krippendorff's α**	**Fleiss' κ**
Social	0.7300	0.7292
Racial	0.7345	0.7338
Type	0.7286	0.7279

Table 9: Inter-annotator agreement across annotation dimensions, showing substantial reliability for all classification tasks.

Step	Dimension	Label	Description	Example
1	Social	Yes	Describes social interaction with pool staff or other clients? **No** → Stop (not social) **Yes** → Proceed to Step 2	*"The receptionist was very friendly and helpful."*
1	Social	No	Describes social interaction with pool staff or other clients? **No** → Stop (not social) **Yes** → Proceed to Step 2	*"The water was too cold and the slides were closed."*
2	Racial	Yes	Social encounter has racial, ethnic, or religious connotations? **No** → 'Other' social encounter **Yes** → Proceed to Step 3	*"There were many migrant families there today."*
2	Racial	No	Social encounter has racial, ethnic, or religious connotations? **No** → 'Other' social encounter **Yes** → Proceed to Step 3	*"The lifeguards were not paying attention to children."*
3	Type	Discriminatory	Uses derogatory language or describes discrimination based on race/ethnicity/religion	*"Too many Arabs who don't know how to behave properly."*
3	Type	Pleasant	Recounts positive experience regarding race/ethnicity/religion	*"No one stared at me for wearing a burkini, very inclusive."*
3	Type	Ambiguous	States fact about race/ethnicity/religion neutrally, without clear sentiment	*"Most staff members speak Turkish and Arabic."*

Table 10: Hierarchical classification framework for review annotation. The three-step process ensures systematic identification of racialised discourse while maintaining contextual sensitivity.

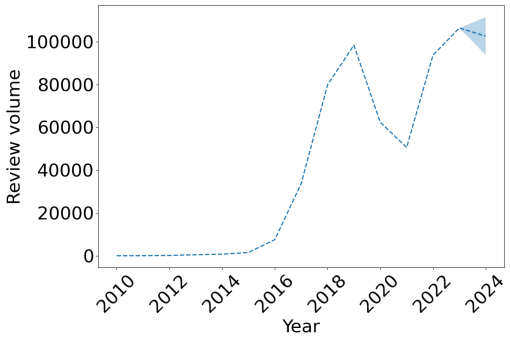


Figure 6: Plausible review volume over time (uncertainty-aware)

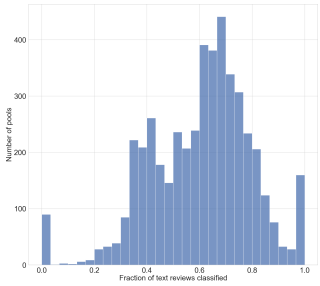


Figure 7: Caption